PHYS 143 – Mechanics and Thermal Physics

Week 4: Work and Kinetic Energy – Supplementary Notes

Hope Donglo, Office 24, Department of Physics, University of Ghana

These notes support the Week 4 lecture. They hold background that did not fit into the contact time. You do not need them to solve the lecture problems, but they show where the results come from.

1 The dot product, briefly

For vectors $\vec{A}$ and $\vec{B}$ with angle ϕ between them (tail to tail), $\vec{A}\cdot\vec{B} = AB\cos\phi = A_xB_x + A_yB_y + A_zB_z$. It is positive for $\phi < 90°$, zero for $\phi = 90°$ and negative for $\phi > 90°$. For work, $W = \vec{F}\cdot\vec{s}$.

2 A short calculus refresher

$\int x^n\,\mathrm{d}x = \dfrac{x^{n+1}}{n+1}$ for $n \neq -1$. Definite integral: $\int_a^b x^n\,\mathrm{d}x = \left[\dfrac{x^{n+1}}{n+1}\right]_a^b$. Example: $\int_0^2 3x^2\,\mathrm{d}x = [x^3]_0^2 = 8$. For a constant factor, $\int c\,f\,\mathrm{d}x = c\int f\,\mathrm{d}x$.

3 The work–energy theorem for a varying force

For motion along x with a force $F_x(x)$, Newton's second law gives $F_x = ma_x = m\,\dfrac{\mathrm{d}v_x}{\mathrm{d}t}$. Since $\mathrm{d}x = v_x\,\mathrm{d}t$,

$$W_{\text{tot}} = \int_{x_1}^{x_2} F_x\,\mathrm{d}x = \int_{x_1}^{x_2} m\frac{\mathrm{d}v_x}{\mathrm{d}t}\,\mathrm{d}x = \int_{v_1}^{v_2} m\,v_x\,\mathrm{d}v_x = \tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2.$$

This shows the theorem holds whatever the force, which is why we could use it with a spring.

4 Work along a general path

For a force that varies along a curved path, the work is the line integral

$$W = \int_{\text{path}} \vec{F}\cdot\mathrm{d}\vec{r},$$

the sum over tiny displacements $\mathrm{d}\vec{r}$ of $F\,\mathrm{d}r\cos\phi$ with the angle ϕ changing along the path. A force perpendicular to $\mathrm{d}\vec{r}$ at every point (for example, tension in a pendulum string) contributes zero.

Why Example 9 (the swing) needs no integral. You pull a child of mass m on a swing of length L slowly, with a horizontal force F, until the chains make an angle θ_0 with the vertical. At each angle θ the child is in equilibrium, so the three forces form a closed triangle and $F = mg\tan\theta$. The child's horizontal displacement for a small change $\mathrm{d}\theta$ is $\mathrm{d}x = L\cos\theta\,\mathrm{d}\theta$, so

$$W_F = \int F\,\mathrm{d}x = \int_0^{\theta_0} mg\tan\theta\;L\cos\theta\,\mathrm{d}\theta = mgL\int_0^{\theta_0}\sin\theta\,\mathrm{d}\theta = mgL\,(1-\cos\theta_0).$$

This is exactly mgh with $h = L(1-\cos\theta_0)$, the answer the work–energy theorem gave in one line. The tension does no work, as the general-path formula says.

5 Work done by a Newton's third-law pair

Let object A exert a force $\vec{\boldsymbol{F}}$ on object B while object B exerts $-\vec{\boldsymbol{F}}$ on A. In a small time the objects have displacements $\mathrm{d}\vec{\boldsymbol{r}}_A$ and $\mathrm{d}\vec{\boldsymbol{r}}_B$. The total work of the pair is

$$\mathrm{d}W_A + \mathrm{d}W_B = (-\vec{\boldsymbol{F}}) \cdot \mathrm{d}\vec{\boldsymbol{r}}_A + \vec{\boldsymbol{F}} \cdot \mathrm{d}\vec{\boldsymbol{r}}_B = \vec{\boldsymbol{F}} \cdot \mathrm{d}(\vec{\boldsymbol{r}}_B - \vec{\boldsymbol{r}}_A).$$

It depends only on the *relative* displacement of the two objects. A weightlifter's hands and the barbell move together, so $\mathrm{d}(\vec{\boldsymbol{r}}_B - \vec{\boldsymbol{r}}_A) = 0$ and the two works cancel: $+Fs$ on the hands and $-Fs$ on the barbell. A block sliding on a fixed table has relative displacement s, so the pair of friction forces has total work $-f_k s$, which is the energy that appears as thermal energy of the surfaces.

6 Where does the energy lost to friction go?

Kinetic friction does negative work on a sliding object and reduces its kinetic energy. The energy is not destroyed: it appears as the *thermal energy* of the two surfaces (they warm up). The work–energy theorem as used in this lecture counts only the kinetic energy of the object, so friction appears as negative work. The full accounting, with potential energy and thermal energy, is the subject of the next lecture and of the thermodynamics part of the course.

7 Units and conversions

$1\,\mathrm{J} = 1\,\mathrm{N\,m}$; $1\,\mathrm{W} = 1\,\mathrm{J/s}$; $1\,\mathrm{kW} = 1000\,\mathrm{W}$; $1\,\mathrm{hp} = 746\,\mathrm{W}$; $1\,\mathrm{kWh} = 3.6 \times 10^6\,\mathrm{J}$. A person climbing stairs develops roughly $10^2\,\mathrm{W}$; a car cruising on a road uses roughly $10^4\,\mathrm{W}$. Household electricity is billed in kilowatt-hours: a $1.2\,\mathrm{kW}$ iron used for $3.0\,\mathrm{h}$ takes $3.6\,\mathrm{kWh}$, which is $1.3 \times 10^7\,\mathrm{J}$.